

Nghia Van Vo | Curriculum Vitae

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Summary

Computational researcher in applied mathematics, optimization, and machine learning with experience designing and analyzing optimization algorithms for large-scale machine learning models. Research focuses on translating mathematical optimization into high-performance implementations, including modern fine-tuning and model-compression techniques.

Education

- Ph.D. in Applied Mathematics, Oakland University, Rochester, Michigan, 2022-present
- M.S. in Applied Statistics, Oakland University, Rochester, Michigan
- Mathematics Teacher Education, University of Education, Ho Chi Minh City, Vietnam, 2017-2021

Publications

- T. A. Nghia Tran, Huy P. N., V. V. H. Khoa, and Nghia Van Vo. Lipschitz stability of regularized linear inverse problems. Preprint, 2026.
- T. A. Nghia Tran, Huy P. N., V. V. H. Khoa, and Nghia Van Vo. Isolated calmness of regularized linear inverse problems. Preprint, 2026.
- P. D. Khanh, B. S. Mordukhovich, D. B. Tran, and Nghia Van Vo. Double proximal point methods for difference-of-convex optimization methods. Preprint, 2026.
- T. A. Nghia Tran, V. V. H. Khoa, and Nghia Van Vo. An effective subspace Newton method for generalized equations. Preprint, 2026.
- P. D. Khanh, V. V. H. Khoa, B. S. Mordukhovich, D. B. Tran, and Nghia Van Vo. Inexact DC algorithms in Hilbert spaces with applications to PDE-constrained optimization. arXiv:2601.06622, submitted to Applied Mathematics & Optimization, 2026.
- H.-C. Luong, Kaiqi Zhao, L. Chen, and Nghia Van Vo. Adaptive temperature scaling for knowledge distillation via entropy-regularized supervision. Submitted, 2026.
- T. A. Nghia Tran, V. V. H. Khoa, and Nghia Van Vo. Nonsmooth Newton methods with effective subspaces for polyhedral regularization. arXiv:2511.16514, submitted to Mathematics of Operations Research, 2025.
- T. A. Nghia Tran, Huy P. N., and Nghia Van Vo. Stable recovery of regularized linear inverse problems. Inverse Problems 41, 065018. DOI: 10.1088/1361-6420/addffb, 2025.
- C.-K. Doan, G.-B. Nguyen, and Nghia Van Vo. A regularity result via fractional maximal operators for p-Laplace equations in weighted Lorentz spaces. Complex Variables and Elliptic Equations. DOI: 10.1080/17476933.2021.1897794, 2022.
- T.-N. Nguyen, M.-P. Tran, C.-K. Doan, and Nghia Van Vo. A gradient estimate related fractional maximal operators for a p-Laplace problem in Morrey spaces. Taiwanese Journal of Mathematics, advance publication, 1-21. DOI: 10.11650/tjm/210202, 2021.

Projects

- Accelerated Image Processing and Optimization: implemented proximal gradient, stochastic subgradient, ADMM, and coordinate descent methods for regression, image restoration, and matrix completion.
- Model Compression using Adaptive Knowledge Distillation: developed Adaptive Temperature Scaling for smaller neural networks with improved accuracy under resource constraints.
- Premier League Ranking Prediction and Disease Symptom Diagnosis: built statistical and machine learning models for structured and unstructured datasets.

Teaching Experience

- MTH 0661 - Elementary Algebra, Main Instructor, Oakland University, Fall 2022, Winter 2023, Winter 2024
- MTH 0662 - Intermediate Algebra, Main Instructor, Oakland University, Fall 2023, Winter 2023
- MTH 1441 - Pre-Calculus, Main Instructor, Oakland University, Fall 2024
- MTH 2554 - Multivariable Calculus, Substitute Instructor, Oakland University, Fall 2025
- MOR 5555 - Nonlinear Optimization, Guest Instructor, Oakland University, Fall 2025

Honors and Awards

- McKay Outstanding Graduate Research Fellowship, Oakland University, 2026
- Travel grants to Midwest Optimization Meeting and East Coast Optimization Meeting, 2023-2025
- Award for Teaching Excellence Internship, Department of Mathematics, Binh Phu High School, 2021
- First Prize, undergraduate scientific research competitions, H.C.M City University of Education, 2020

Skills

- Programming: Python, Matlab, SAS, C, SQL
- Optimization: gradient descent, stochastic gradient descent, proximal methods, FISTA, Newton methods, ADMM, Douglas-Rachford, Chambolle-Pock primal-dual, coordinate descent
- Tools: Scikit-learn, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, CVXPY, Gurobi, CPLEX, MATLAB Optimization Toolbox, Git, GitHub
- Languages: English, Vietnamese